BITS Pilani
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Introduction to Statistical Methods

Course code: DSECL ZC418 / AIML CZC418

ISM TEAM

AIMLCZC418, ISM
Lecture No. 15

AGENDA

- What is a Gaussian Mixture Model?
- Mathematical Representation
- Understanding Components of GMM
- Implementation Steps

Why does it work?

Needs a fair bit of math to understand it

What is a Gaussian Mixture Model?

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- A Gaussian Mixture Model is a probabilistic model that represents a mixture of Gaussian (normal) distributions. Each Gaussian component represents one cluster within the data.

$$\frac{1}{|\Sigma|^{-\frac{1}{2}}(2\pi)^{D/2}} e^{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)}$$

- It assumes that the data is generated from a mixture of several Gaussian distributions with unknown parameters.

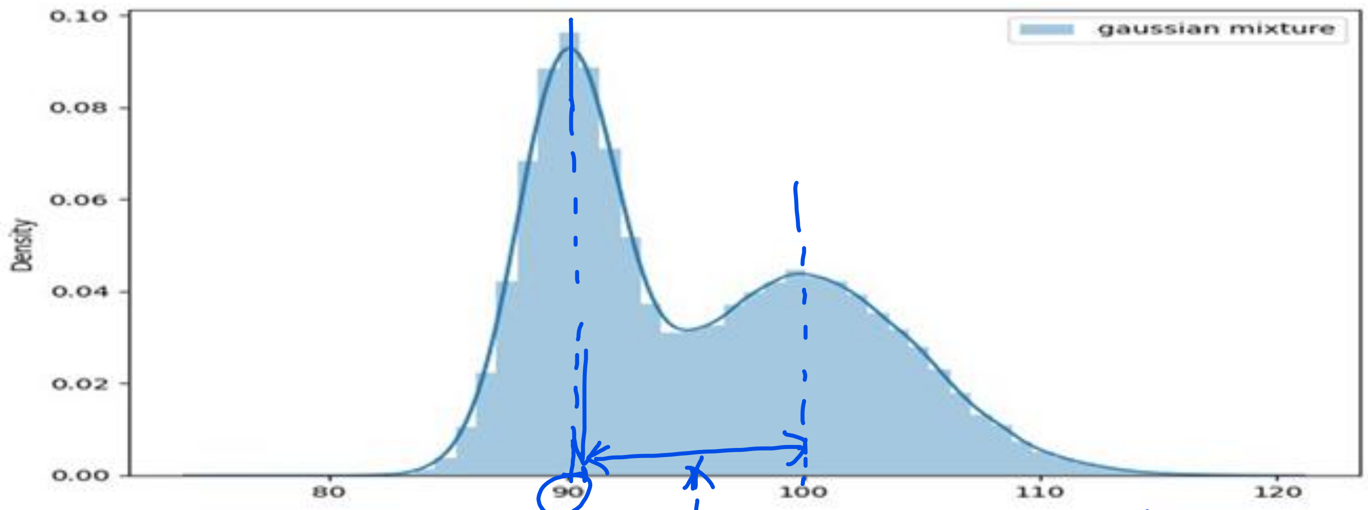
- GMM is a soft clustering algorithm, meaning it assigns a probability to each point to belong to each cluster.

- Note: Hard clustering : Clusters do not overlap

  Soft clustering : Clusters may overlap

Gaussian Mixture Model

- Suppose Company A share price is normally distributed with mean price Rs. 90 and standard deviation of price Rs.2 with 1000 sample points
- Company B share price is normally distributed with mean price Rs. 100 and standard deviation of price Rs.2 with 800 sample points
- Now, both samples are mixed, then we obtain Normal (Gaussian) mixture model.



→ a point here could have been generated from the 1st dist or the 2nd dist

So, after mixing the processes together, we have the dataset that we see on the plot. We can notice 2 peaks: around 90 and 100, but for many of the points in the middle of the peaks it is ambiguous to which distribution they were drawn from. So how should we approach this problem?

- Gaussian Mixture Models (GMMs) are a popular and powerful technique in statistics and machine learning for modeling complex data distributions.

- They are particularly useful for clustering and density estimation.

Mathematical Representation

Given a dataset X = $\{x_1, x_2, \ldots, x_n\}$, where x_i is a d-dimensional feature vector. The probability density function of a GMM is represented as:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

Mean & covariance matrix of the kth distribution

$\Sigma \pi_k = 1$

where:

π_k is the mixing coefficient for component k (sums to 1)

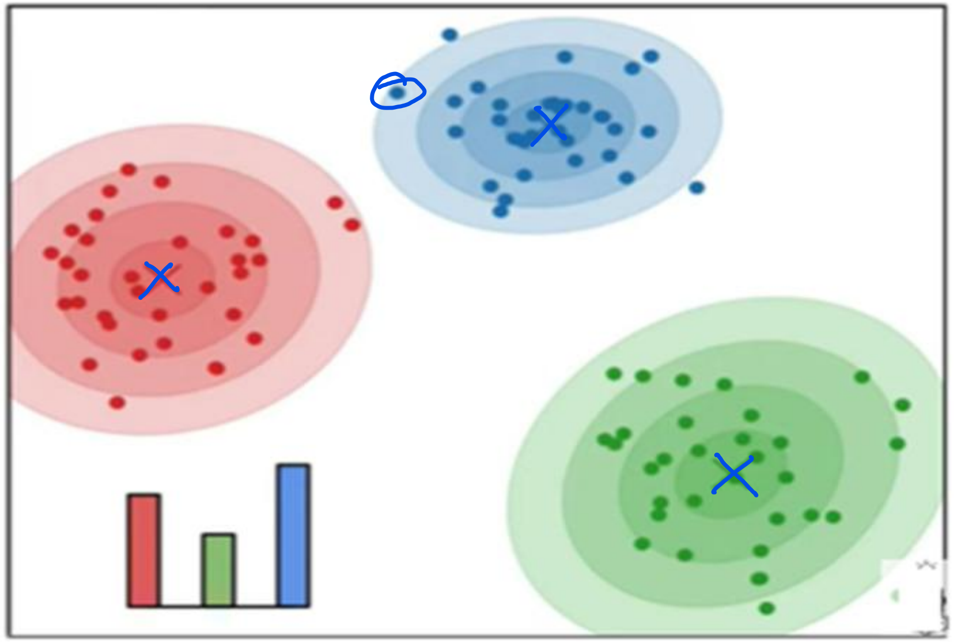
$\mathcal{N}(x \mid \mu_k, \Sigma_k)$represents the multivariate Gaussian distribution with mean μ_k and covariance matrix Σ_k

Understanding Components of GMM

- Mean ($\boldsymbol{\mu_k}$): Represents the center of each Gaussian component.
- Covariance ($\boldsymbol{\Sigma_k}$): Describes the shape and orientation of the data distribution.
- Mixing Coefficients ($\boldsymbol{\pi_k}$): Represents the weight of each Gaussian component in the mixture.
- Number of Components ($\boldsymbol{K}$): Determines the number of Gaussian distributions in the mixture.
- These parameters determine the shape, location, and importance of each Gaussian component.

Understanding Components of GMM

1. The Mean



2. The Covariance Matrix

3. The Weight

Implementation Steps

- Initialization: Initialize the parameters (mean, covariance, mixing coefficients).
- Expectation-Maximization (EM) Algorithm:
  - E-step: Compute the posterior probability of each component (Expected value of the likelihood function) using the current parameters.
  - M-step: Update the parameters(means, covariances, and mixing coefficients) to maximize the likelihood of the data.
- Convergence: Repeat the EM steps until convergence criteria are met.
- Predictions: Assign data points to the most probable cluster.

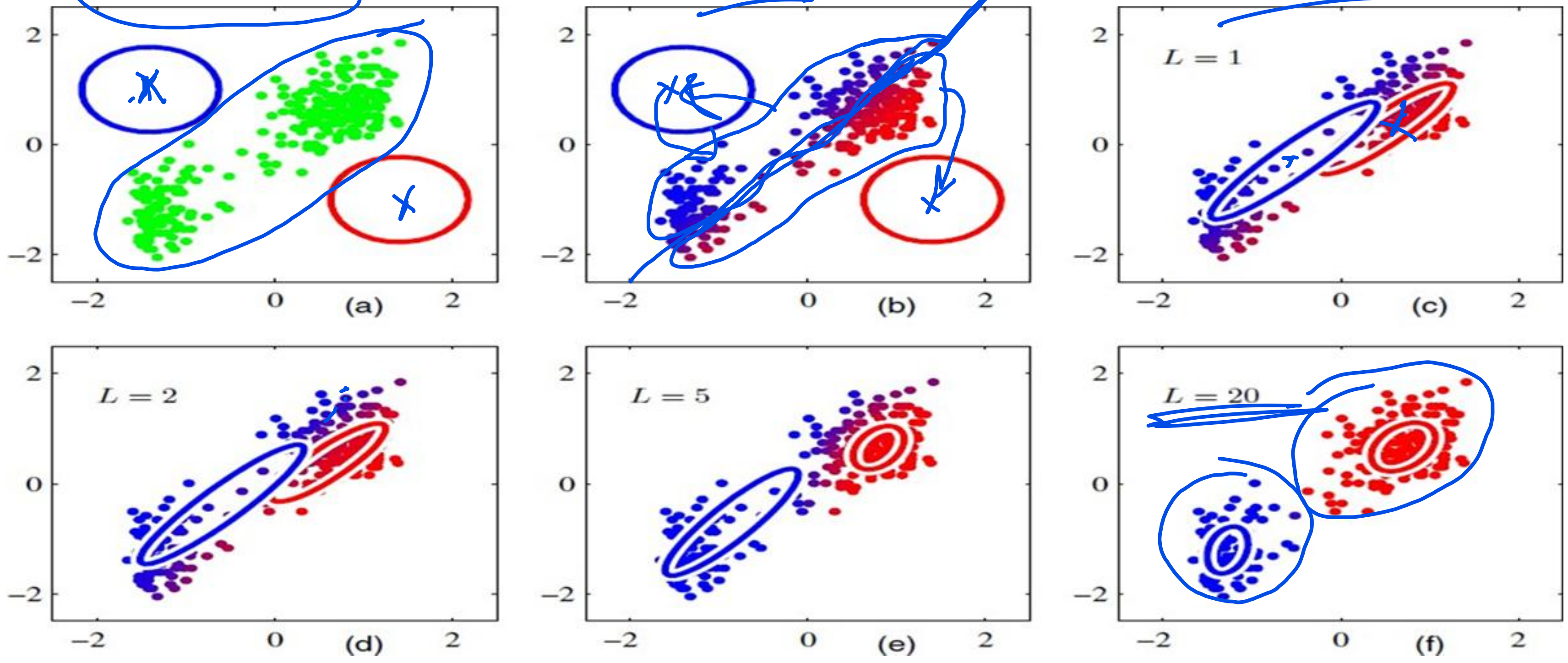
Expectation Maximization (EM) Algorithm

- We first choose some initial values for the means, covariances, and mixing coefficients.
- Then we alternate between the following two updates that we shall call the E step and the M step
- In the expectation step, or E step, we use the current values for the parameters to evaluate the posterior probabilities,
- We then use these probabilities in the maximization step, or M step, to re-estimate the means, covariances, and mixing co-eff.
- In practice, the algorithm is deemed to have converged when the change in the log likelihood function, or alternatively in the parameters, falls below some threshold

EM Algorithm

evaluate responsibilities

Initialization E Step M Step



Challenges

1. GMMs assume that data is generated from a mixture of Gaussian distributions, which may not hold for all types of data.

2. The number of components in the GMM must often be chosen beforehand, which can be a challenging task.

Applications of GMM

- **Clustering:** GMMs can be used for soft clustering, where each data point belongs to multiple clusters with associated probabilities.

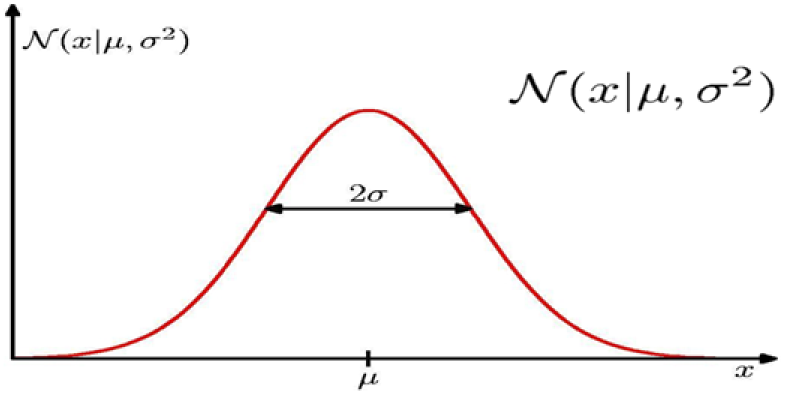
- **Density Estimation:** GMMs can estimate the underlying probability distribution of data.

- **Anomaly Detection:** Unusual data points can be detected by looking at their likelihood under the GMM.

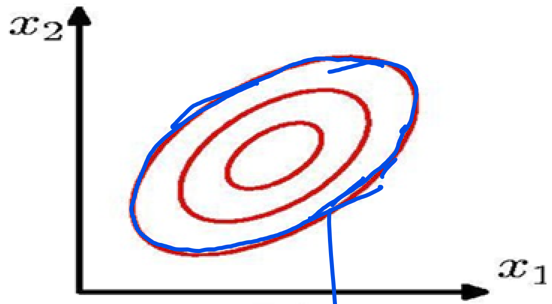
Gaussian Distribution

$$\Sigma = \begin{bmatrix} \sigma_1^2 & r\sigma_1\sigma_2 \\ r\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}_{2\times 2}$$

correlation coefficient



$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$$



Multivariate Gaussian Distribution

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}$$

$N(x|\mu,\Sigma)$ is the same on all points of this contour line

(Ref: Chris Bishop)

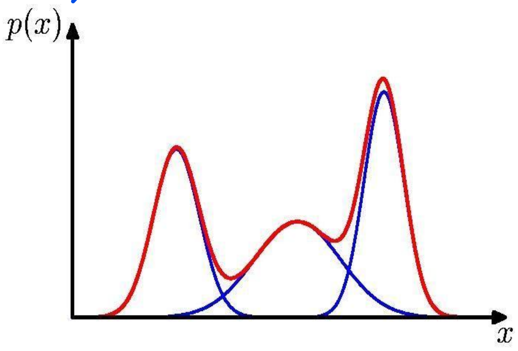
Mixtures of Gaussian

$p(x) = p(x =$

Combine simple models into a complex model:

$p(x, z=k)$

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \underbrace{\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}_{\text{Component}}$$



Component

Mixing coefficient

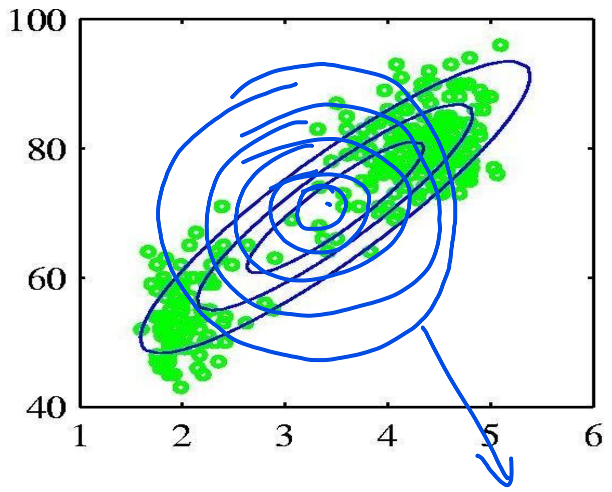
$$\forall k : \pi_k \geqslant 0 \qquad \sum_{k=1}^{K} \pi_k = 1$$

$$\int_{-\infty}^{\infty} p(x)dx \stackrel{?}{=} 1$$

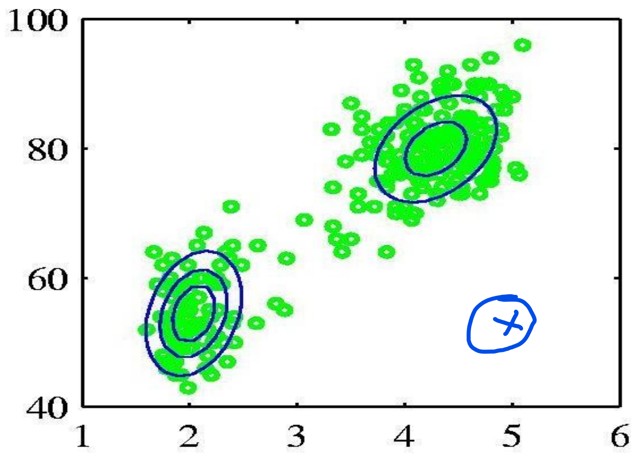
Find parameters through EM (Expectation Maximization) algorithm

Ref: Adapted from Chris Bishop

Probabilistic version: Mixtures of Gaussians



Single Gaussians



Mixture of 2 Gaussians

GMM

l_1

- Data set: $D = \{x_n\}, n = 1, \cdots, N$
- Consider first a single Gaussian
- Assume observed data points generated independently
  - $p(D|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \prod_{n=1}^{N} N(x_n|\boldsymbol{\mu}, \boldsymbol{\Sigma}\}$
- Viewed as a function of the parameters, this is known as the *likelihood function*

$z_i = 3$

GMM

$z = [0\ 0\ 1\ 0\ 0\ 0]$ choosing 3rd Gaussian

- K-dimensional binary random variable z having a 1-of-K representation in which a particular element z_k is equal to 1 and all other elements are equal to 0.
- The values of z_k therefore satisfy $z_k \in \{0,1\}$
- K possible states for the vector z according to which element is nonzero.
- Joint distribution p(x,z) in terms of a marginal distribution p(z) and a conditional distribution p(x|z),
- Marginal distribution over z is specified in terms of the mixing coefficients π_k, such that $p(z_k = 1) = \pi_k$

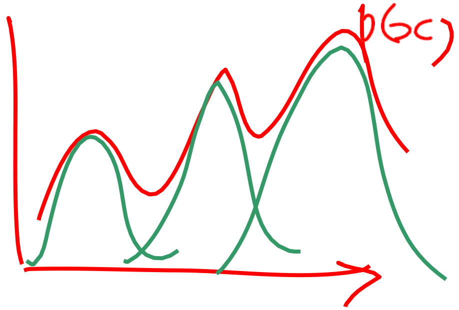
Fitting the Gaussian Mixture Model

- We wish to invert this process – given the data set, find the corresponding parameters:
  - mixing coefficients
  - means
  - covariances
- If we knew which component generated each data point, the maximum likelihood solution would involve fitting each component to the corresponding cluster
- Problem: the data set is unlabelled
- We shall refer to the labels as *latent* (= hidden) variables

GMM

- Linear super-position of Gaussians

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$



- Normalization and positivity require

$$\sum_{k=1}^{K} \pi_k = 1 \qquad 0 \leqslant \pi_k \leqslant 1$$

- Can interpret the mixing coefficients as prior probabilities

$$p(\mathbf{x}) = \sum_{k=1}^{K} p(k)\, p(\mathbf{x}|k)$$

$\mathcal{N}(x|\mu_k, \Sigma_k)$

GMM

- z uses a 1-of-K representation, we can also write this distribution in the form

$$p(\mathbf{z}) = \prod_{k=1}^{K} \pi_k^{z_k}$$

$z = [0 \ 0 \dots 1 \ 0 \ 0]$ $\quad \to 5$

$\prod_{k=1}^{K} \pi_k^{z_k} = \pi_1^0 \, \pi_2^0 \, \pi_3^0 \cdots \pi_5^1 \, \pi_6^0 \, \pi_7^0$

$= \pi_5^1$

- Conditional distribution of x given a particular value for z is a Gaussian

$$p(\mathbf{x}|\mathbf{z}) = \prod_{k=1}^{K} \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_k}$$

$\underbrace{\mathcal{N}(x|\mu_1, \Sigma_1)^0}_{=1} \, \mathcal{N}(x|\mu_2, \Sigma_2)^0 \cdots \mathcal{N}(x|\mu_5, \Sigma_5)^1 \cdots$

- Joint distribution is given by p(z)p(x|z), and the marginal distribution of x is then obtained by summing the joint distribution over all possible states of z to given

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z})p(\mathbf{x}|\mathbf{z}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

GMM

- Conditional probability of z given x
- use $\gamma(z_k)$ to denote $p(z_k = 1 \mid x)$, whose value can be found using Bayes' theorem

$$\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) = \frac{p(z_k = 1)p(\mathbf{x}|z_k = 1)}{\sum_{j=1}^{K} p(z_j = 1)p(\mathbf{x}|z_j = 1)} = \frac{p(x, z_k)}{p(x)}$$

$$p(z_k = 1 / x) = \frac{p(x, z_k)}{p(x)} = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

$$\sum_{j=1}^{K} p(x, z_j) = p(x)$$

- π_k as the prior probability of $z_k = 1$, and the quantity $\gamma(z_k)$ as the corresponding posterior probability once we have observed x.

Maximum Likelihood Function

- Log likelihood function:

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

- Maximizing the log likelihood function for a Gaussian mixture model turns out to be a more complex problem than for the case of a single Gaussian.

- The difficulty arises from the presence of the summation over k that appears inside the logarithm, so that the logarithm function no longer acts directly on the Gaussian.

GMM Problems and Solutions

- How to maximize the log likelihood
  - solved by expectation-maximization (EM) algorithm
- How to avoid singularities in the likelihood function
  - solved by a Bayesian treatment
- How to choose number *K* of components
  - also solved by a Bayesian treatment

EM Algorithm

- Conditions for MLE: Setting the derivatives of
- $\ln p(X|\pi, \mu, \Sigma)$ in with respect to the means μ_k
- of the Gaussian components to zero, we obtain

$$0 = -\sum_{n=1}^{N} \underbrace{\frac{\pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}}_{\gamma(z_{nk})} \boldsymbol{\Sigma}_k(\mathbf{x}_n - \boldsymbol{\mu}_k)$$

- Rearranging, we obtain

$$\boxed{\boldsymbol{\mu}_k = \frac{1}{N_k}\sum_{n=1}^{N}\gamma(z_{nk})\mathbf{x}_n}$$

- Note here we have defined

$$\boxed{N_k = \sum_{n=1}^{N}\gamma(z_{nk})}$$

$\nabla_\pi \ln p(X|\pi, \mu, \Sigma) = 0$

$\nabla_\mu \ln p(X|\pi, \mu, \Sigma) = 0$

$\nabla_\Sigma \ln p(X|\pi, \mu, \Sigma) = 0$

$\gamma(z_{nk}) = \dfrac{p(X = x_n, z_k = 1)}{\sum_j p(X = x_n, z_j = 1)}$

optimization

stationary point

total responsibility associated with Gaussian k

EM Algorithm

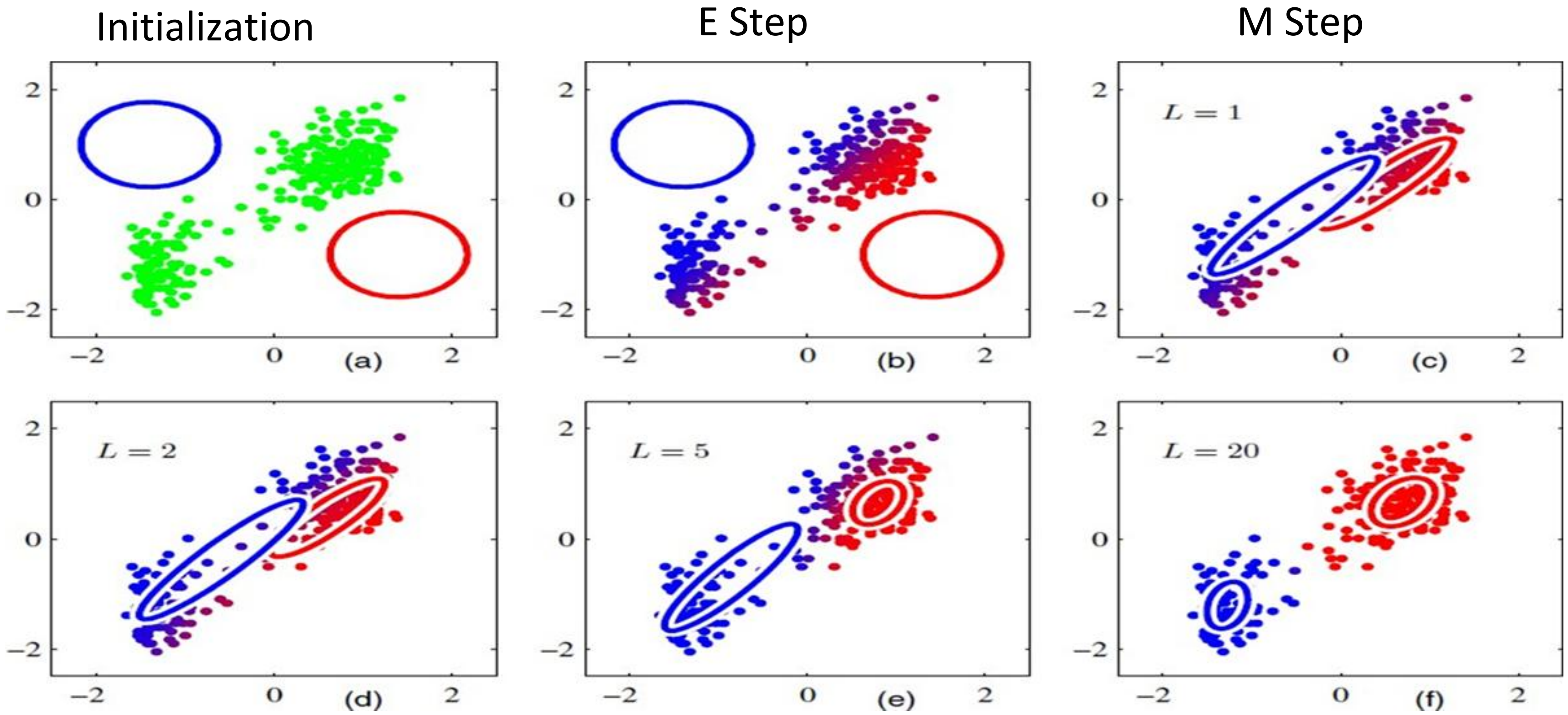
- We first choose some initial values for the means, covariances, and mixing coefficients.
- Then we alternate between the following two updates that we shall call the E step and the M step
- In the expectation step, or E step, we use the current values for the parameters to evaluate the posterior probabilities, ($\gamma(z_{nk})$)
- We then use these probabilities in the maximization step, or M step, to re-estimate the means, covariances, and mixing
- In practice, the algorithm is deemed to have converged when the change in the log likelihood function, or alternatively in the parameters, falls below some threshold

EM Algorithm : Informal Derivation

- The solutions are not closed form since they are coupled
- Suggests an iterative scheme for solving them:
  - make initial guesses for the parameters
  - alternate between the following two stages:
    - E-step: evaluate responsibilities
    - M-step: update parameters using ML results
- Each EM cycle guaranteed not to decrease the likelihood

we do not prove it here → will give hint of proof if time permits

EM Algorithm



EM Algorithm for GMM

1. Initialize the means μ_k, covariances Σ_k and mixing coefficients π_k, and evaluate the initial value of the log likelihood.

2. **E step:** Evaluate the responsibilities using the current parameter values

$$\gamma(z_{nk}) = \frac{\pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

responsibilities

$p(z_k = 1 / x_n)$

$= \gamma(z_{nk})$

EM Algorithm for GMM

3. **M step:** Re-estimate the parameters using the current responsibilities

$$\frac{d \ln p(X \mid \pi, \mu, \Sigma)}{d\mu,\ d\pi,\ d\Sigma} = 0$$

$$\boldsymbol{\mu}_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) \mathbf{x}_n$$

weighted center of mass

$$\boldsymbol{\Sigma}_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) \left(\mathbf{x}_n - \boldsymbol{\mu}_k^{\text{new}}\right)\left(\mathbf{x}_n - \boldsymbol{\mu}_k^{\text{new}}\right)^{\mathrm{T}}$$

weighted covariance matrix

$$\pi_k^{\text{new}} = \frac{N_k}{N}$$

where

$$N_k = \sum_{n=1}^{N} \gamma(z_{nk})$$

= total responsibility of the kth Gaussian / total # of point

EM Algorithm for GMM

4. Evaluate the log likelihood

$$\ln p(\mathbf{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

Problem 1

A dataset consists of one-dimensional observations that appear to come from **two different Gaussian distributions**. The probability density function is modelled as a Gaussian Mixture Model:

$$p(x) = \pi_1 \mathcal{N}(x|\mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(x|\mu_2, \sigma_2^2)$$

where
$\pi_1 = 0.6,\ \pi_2 = 0.4,$
$\mu_1 = 90,\ \sigma_1 = 5,$
$\mu_2 = 100,\ \sigma_2 = 5.$

Problem 1 (contd.)

1. Write the explicit expression for $p(x)$.

2. For a data point $x = 95$, compute the **responsibility** (posterior probability) that it belongs to the first Gaussian component.

$$\gamma_1(x) = \frac{\pi_1 \mathcal{N}(x|\mu_1, \sigma_1^2)}{p(x)}$$

Solution

The Gaussian Mixture Model is

$$p(x) = \pi_1 \mathcal{N}(x|\mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(x|\mu_2, \sigma_2^2)$$

with parameters:

$$\pi_1 = 0.6, \pi_2 = 0.4$$
$$\mu_1 = 90, \mu_2 = 100$$
$$\sigma_1 = \sigma_2 = 5 \ (\sigma^2 = 25)$$

Explicit expression for $p(x)$

$$p(x) = \frac{0.6}{\sqrt{2\pi}5} exp\left(-\frac{(x-90)^2}{50}\right) + \frac{0.4}{\sqrt{2\pi}5} exp(-\frac{(x-100)^2}{50})$$

Solution

b) Responsibility of component 1for $x = 95$

$$\gamma_1 = \frac{\pi_1 N(95|90,25)}{\pi_1 N(95|90,25) + \pi_2 N(95|100,25)}$$

$$= \frac{0.6}{0.6 + 0.4} = 0.6$$

Problem 2

A dataset consists of one-dimensional observations modeled using a **two-component Gaussian Mixture Model**:

$$p(x) = \pi_1 \mathcal{N}(x|\mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(x|\mu_2, \sigma_2^2)$$

Initial parameters are given as:

$$\pi_1^{(0)} = 0.6, \pi_2^{(0)} = 0.4$$

$$\mu_1^{(0)} = 90, \mu_2^{(0)} = 100$$

$$\sigma_1^{(0)} = \sigma_2^{(0)} = 5$$

The observed data points are:

$$x = \{92, 95, 98\}$$

Problem 2 (contd.)

(a) E-step: For each data point x_i ,compute the **responsibility** of component

(b) M-step: Using the responsibilities computed in part (a), update the mixing coefficients and hence the mean of the first component

(only for 1 iteration)

Solution

Two-component GMM with initial parameters:

$$\pi_1^{(0)} = 0.6, \pi_2^{(0)} = 0.4$$
$$\mu_1^{(0)} = 90, \mu_2^{(0)} = 100$$
$$\sigma_1 = \sigma_2 = 5 \;\Rightarrow\; \sigma^2 = 25$$

Data points:

$$x = \{92, 95, 98\}, N = 3$$

E-step: $\gamma_{i1} = \frac{\pi_1 N(x_i|\mu_1,25)}{\pi_1 N(x_i|\mu_1,25)+\pi_2 N(x_i|\mu_2, 25)}$

$\Rightarrow \gamma_{11} = (0.6e^{-0.08})/(0.6e^{-0.08} + 0.4e^{-1.28})$=0.83

Solution

Similarly $\gamma_{21} = 0.6, \gamma_{31} = 0.31$

b) M Step

Mixing coefficients

$$\pi_1^{(1)} = \left(\frac{1}{3}\right)(0.83 + 0.6 + 0.31) = 0.58$$

$$\pi_2^{(1)} = \left(1 - \pi_1^{(1)}\right) = 0.42$$

Mean of the 1st component

$$\mu_1^{(1)} = \frac{\sum_{i=1}^{3} \gamma_{i1} x_n}{\sum_{n=1}^{3} \gamma_{i1}}$$

=[0.83(0.92)+0.6(0.95)+0.31(0.98)]/(0.83+0.6+0.31)

=94.1

Thank you ☺